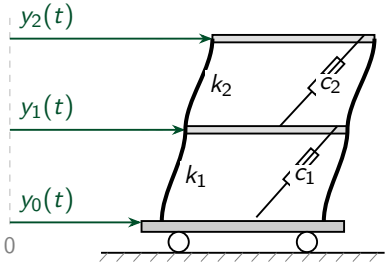
Two-Story Building

Displacements: $\boldsymbol{y}(t) = (y_1(t), y_2(t))$ (m), t in seconds.



$m_2 = 3\,\mathrm{kg}$
$k_2 = 3\,\mathrm{N/m}$
$c_2 = 1\,\mathrm{N\,s/m}$

$m_1 = 4\,\mathrm{kg}$
$k_1 = 4\,\mathrm{N/m}$
$c_1 = 2\,\mathrm{N\,s/m}$

$y_0(t) = \sin(t)\ \mathrm{m}$

Worked problem

The base motion $y_0(t)$ is prescribed. The two floor displacements are measured to the right from rest.

$$\boldsymbol{y}(0) = (0, -1)\,\mathrm{m}$$
$$\boldsymbol{y}'(0) = (0, 2)\,\mathrm{m/s}$$

Story Forces and Floor Balances

Story i connects levels $i-1$ and i. The force on the upper level is $-f_i$ and the force on the lower level is $+f_i$, where

$$f_i = k_i(y_i - y_{i-1}) + c_i(y_i' - y_{i-1}')$$

$$\underbrace{\frac{\mathrm{N}}{\cancel{\mathrm{m}}}\cancel{\mathrm{m}} = \mathrm{N}}_{\text{spring}} \qquad \underbrace{\frac{\mathrm{N}\,\cancel{\mathrm{s}}}{\cancel{\mathrm{m}}}\frac{\cancel{\mathrm{m}}}{\cancel{\mathrm{s}}} = \mathrm{N}}_{\text{damper}} \qquad \mathrm{kg}\,\frac{\mathrm{m}}{\mathrm{s}^2} = \mathrm{N}$$

Substituting the story forces into Newton's law gives

$$\begin{aligned} 4y_1'' = -f_1 + f_2 \quad &= -4(y_1 - y_0) - 2(y_1' - y_0') + 3(y_2 - y_1) + (y_2' - y_1') \\ 3y_2'' = -f_2 \quad &= \qquad\qquad - 3(y_2 - y_1) - (y_2' - y_1') \end{aligned}$$

Step-by-step algebra

Floor displacements are blue, velocities are red, and accelerations are green.

$$\begin{aligned} 4y_1'' &= -4(y_1 - y_0) - 2(y_1' - y_0') + 3(y_2 - y_1) + (y_2' - y_1') \\ 3y_2'' &= \qquad\qquad\qquad\qquad\quad - 3(y_2 - y_1) - (y_2' - y_1') \end{aligned}$$

Move every floor term to the left. Keep the prescribed base terms on the right.

$$\begin{aligned} 4y_1'' + 2y_1' - (y_2' - y_1') + 4y_1 - 3(y_2 - y_1) &= 4y_0 + 2y_0' \\ 3y_2'' + (y_2' - y_1') + 3(y_2 - y_1) &= 0 \end{aligned}$$

Collect like terms in each floor equation.

$$\begin{aligned} 4y_1'' + 3y_1' - y_2' + 7y_1 - 3y_2 &= 4y_0 + 2y_0' \\ 3y_2'' - y_1' \quad + y_2' - 3y_1 + 3y_2 &= 0 \end{aligned}$$

Floor Balances in Matrix Form

Collect the floor terms on the left and the prescribed base motion on the right.

$$\underbrace{\begin{pmatrix} 4 & 0 \\ 0 & 3 \end{pmatrix}}_{\boldsymbol{M}} \begin{pmatrix} y_1'' \\ y_2'' \end{pmatrix} + \underbrace{\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}}_{\boldsymbol{C}} \begin{pmatrix} y_1' \\ y_2' \end{pmatrix} + \underbrace{\begin{pmatrix} 7 & -3 \\ -3 & 3 \end{pmatrix}}_{\boldsymbol{K}} \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \underbrace{\begin{pmatrix} 4y_0 + 2y_0' \\ 0 \end{pmatrix}}_{\boldsymbol{b}(t)}$$

$$\boldsymbol{M}\boldsymbol{y}'' + \boldsymbol{C}\boldsymbol{y}' + \boldsymbol{K}\boldsymbol{y} = \boldsymbol{b}(t) \qquad \boldsymbol{b}(t) = \begin{pmatrix} 4y_0(t) + 2y_0'(t) \\ 0 \end{pmatrix}$$

$\boldsymbol{M}$ is in kg. $\boldsymbol{C}$ is in $\mathrm{N\,s/m}$. $\boldsymbol{K}$ is in $\mathrm{N/m}$. $\boldsymbol{b}$ is in N. Base motion is $y_0(t) = \sin(t)\ \mathrm{m}$.

First-Order version $z = (y_1, y_2, y_1', y_2') = (y_1, y_2, v_1, v_2)$

The equations $y_1' = v_1$ and $y_2' = v_2$ give the top two rows of the system.

$$\begin{pmatrix} y_1' \\ y_2' \end{pmatrix} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \boldsymbol{I} \end{pmatrix} \boldsymbol{z}$$

Solving the ODEs for $v_1' = y_1''$ and $v_2' = y_2''$ gives the bottom rows of the system.

$$\boldsymbol{z}' = \boldsymbol{A}\boldsymbol{z} + \boldsymbol{g}(t) \qquad \boldsymbol{A} = \begin{pmatrix} \mathbf{0} & \boldsymbol{I} \\ -\boldsymbol{M}^{-1}\boldsymbol{K} & -\boldsymbol{M}^{-1}\boldsymbol{C} \end{pmatrix} \qquad \boldsymbol{g}(t) = \begin{pmatrix} \mathbf{0} \\ \boldsymbol{M}^{-1}\boldsymbol{b}(t) \end{pmatrix}$$

$$\boldsymbol{A} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -7/4 & 3/4 & -3/4 & 1/4 \\ 1 & -1 & 1/3 & -1/3 \end{pmatrix} \quad \boldsymbol{g}(t) = \begin{pmatrix} 0 \\ 0 \\ (4y_0(t) + 2y_0'(t))/4 \\ 0 \end{pmatrix} \quad \boldsymbol{z}(0) = \begin{pmatrix} 0 \\ -1 \\ 0 \\ 2 \end{pmatrix}$$